



COL7378: Principles of Autonomous Systems

Semester II, 2025-26

Agent Representation - I

Rohan Paul

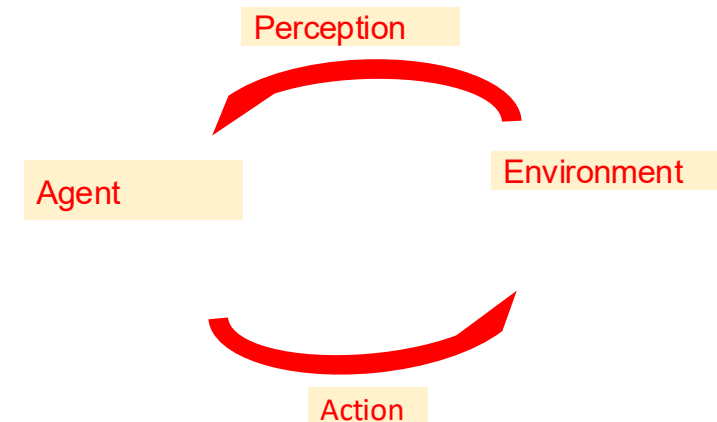


Today's lecture

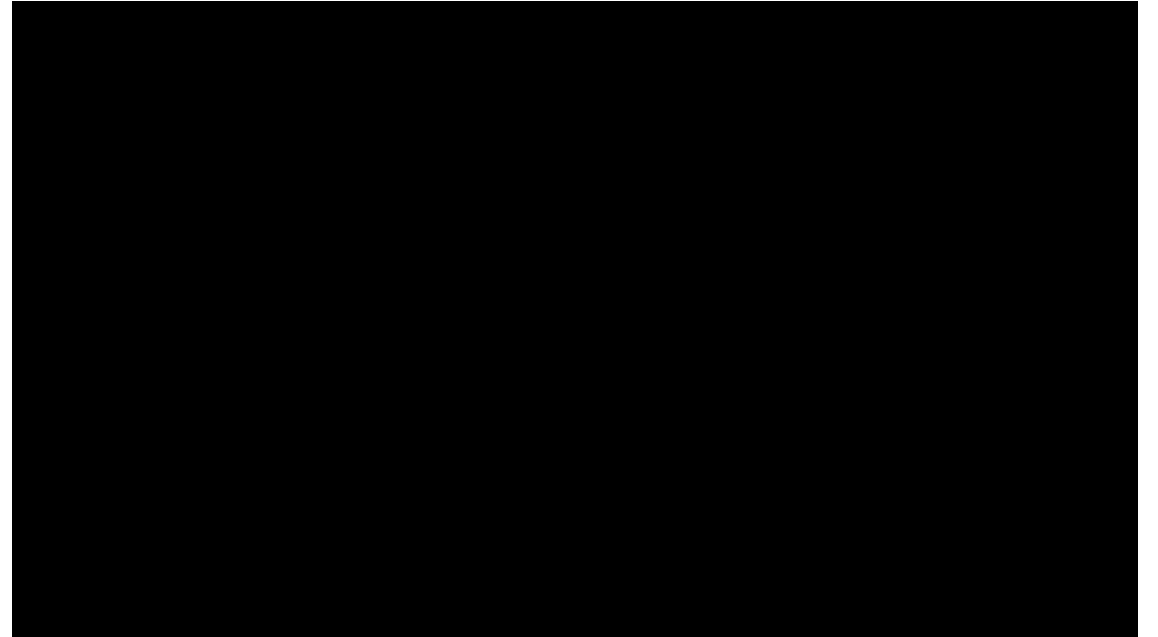
- Last Class
 - Course Introduction
- This Class
 - Agent Representation - I

Agentic View

- Agents act in the world
 - Possess a goal.
 - Takes observations from the environment – form an understanding of the world.
 - Plan actions – reason, execute, correct
- Science of Autonomy
 - What model/algorithms one should have that intelligent behaviour arises naturally by an objective minimization/search.
- Continual Interaction
 - Data over time – on their own or from the human.
 - Hence, can learn from data (ML problem)

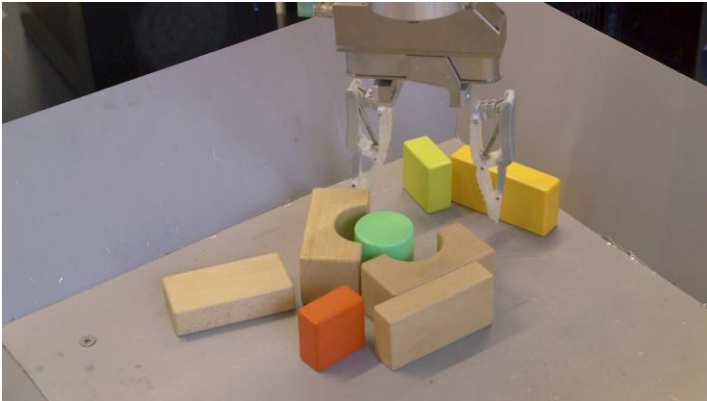


Physical Interactions: Generating Movement



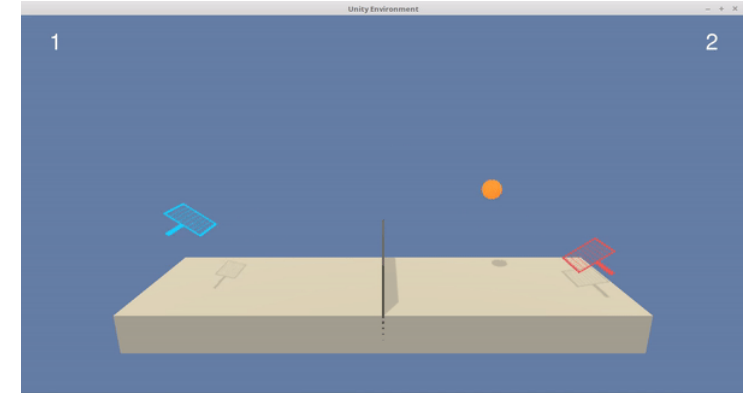
Other agent interactions

- For some environments and tasks a clear separation may not be modeled.



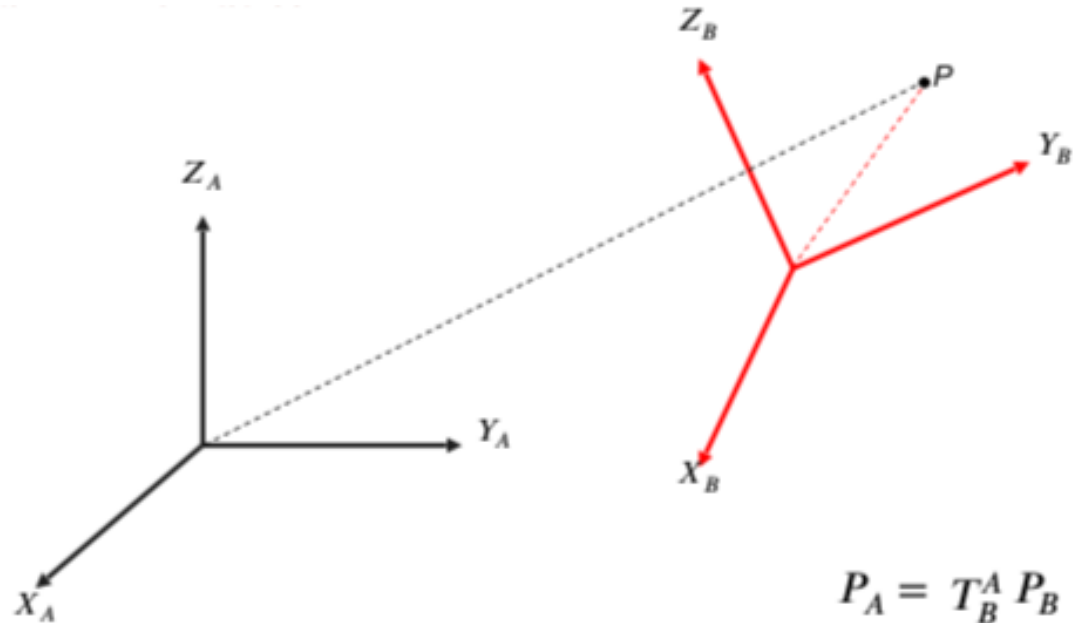
<https://ai.googleblog.com/2016/10/how-robots-can-acquire-new-skills-from.html>

Another example: AI Habitat <https://aihabitat.org/>



Coordinate Frames

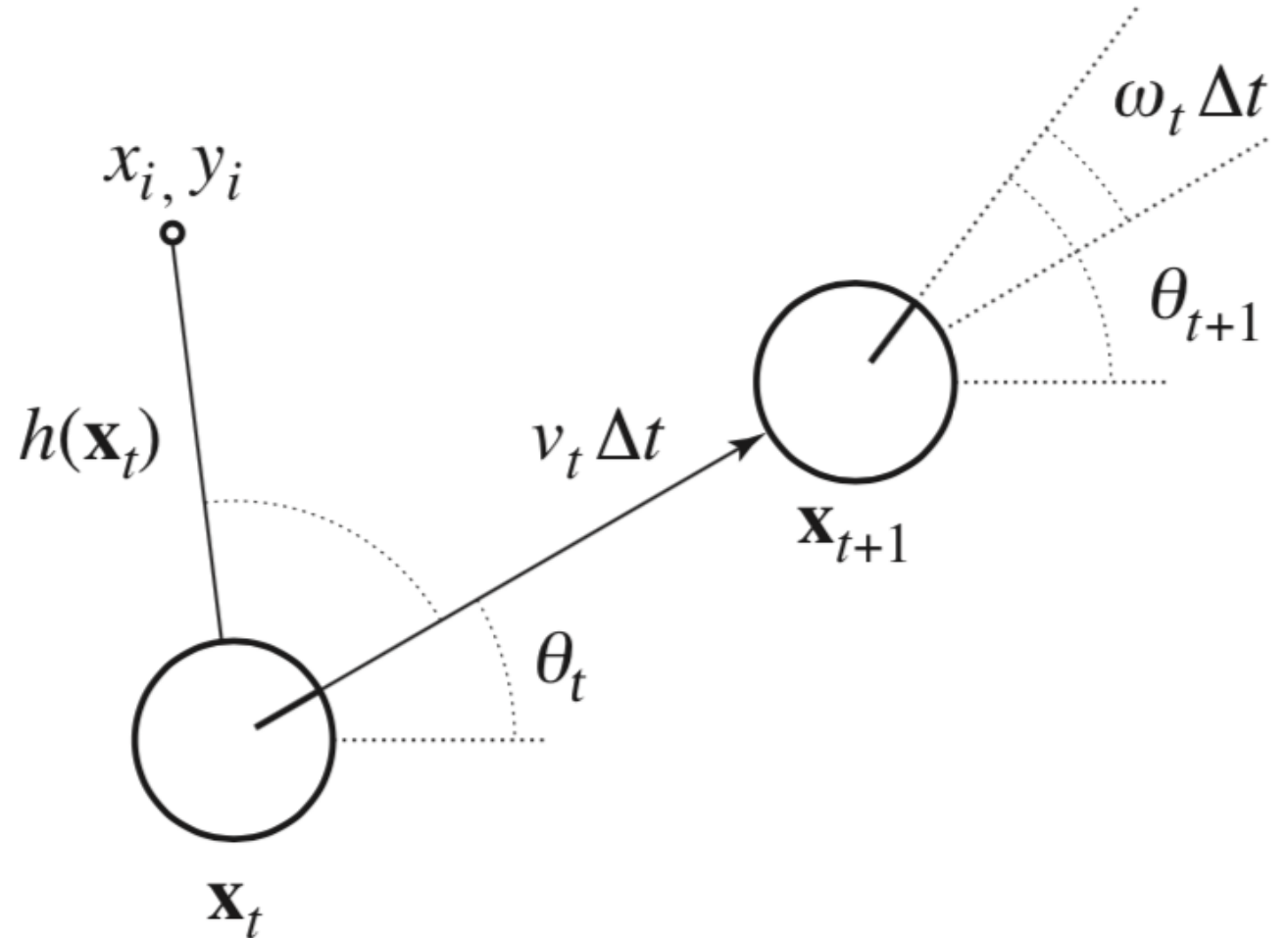
- Coordinate frame
 - Global
 - On the agent.
- Pose, a particular point on its body, (e.g., x, y, z, heading etc.) with respect to a coordinate frame.
- State changes occur due to actions the agent takes.
- Need a way to determine the pose w.r.t. any coordinate frame.
- Relationship maintained in coordinate transforms.



$$P_A = T_B^A P_B$$
$$T_B^A = \begin{bmatrix} r_{11} & r_{21} & r_{31} & \Delta x \\ r_{12} & r_{22} & r_{32} & \Delta y \\ r_{13} & r_{23} & r_{33} & \Delta z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

How does movement occur?

- Simple agents
 - Point-like or disk-like agents
- Linear and angular velocities change the state from current to the next time step.
- Position and heading direction is changed.



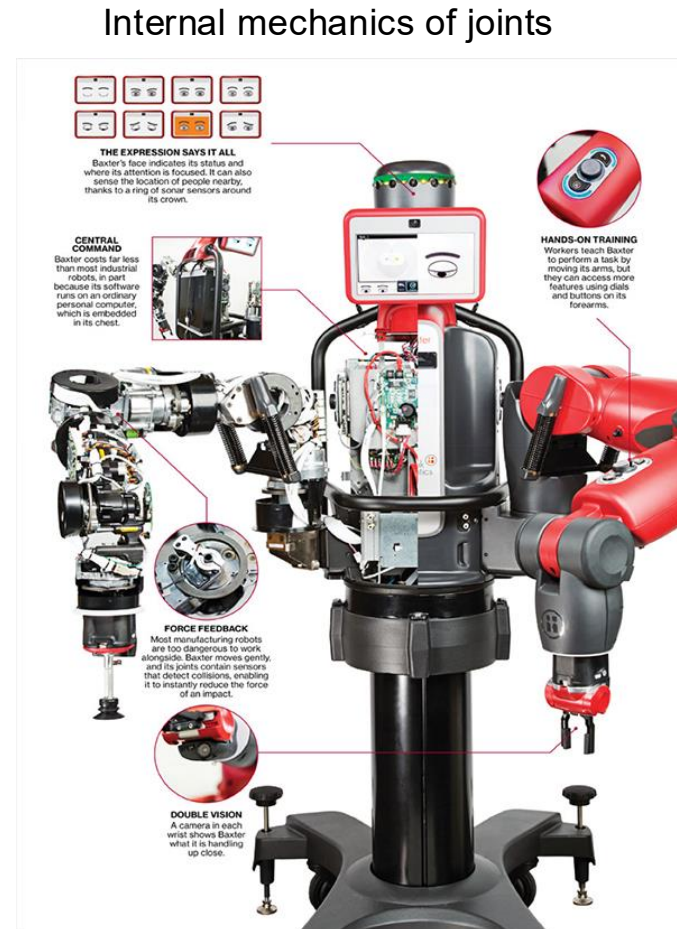
More complex physical agents

- An agent capable of manipulation
 - Simply a number of links connected via joints.

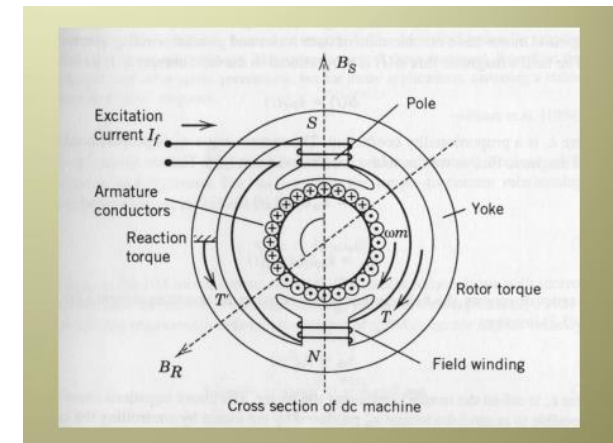


How is movement generated ?

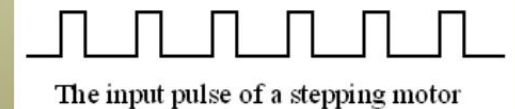
- Joints typically have motors that exert torque.
- Also called actuation
Typically, abstracted away for algorithmic purposes.



DC motor

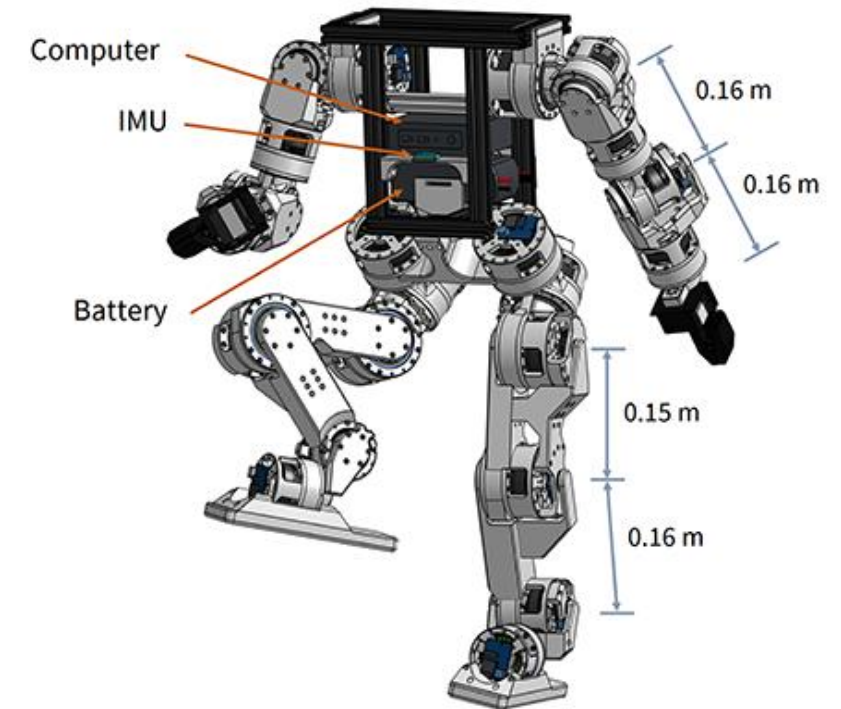
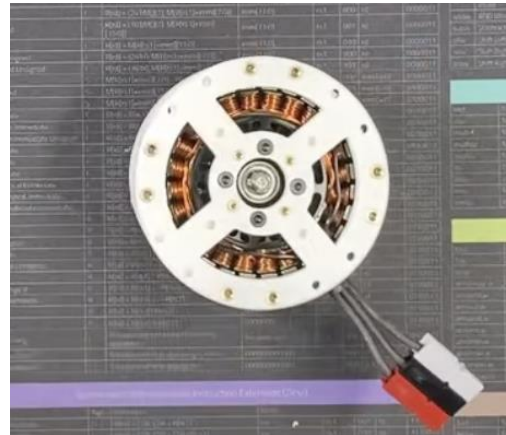


Stepper motor



What causes an arm to move?

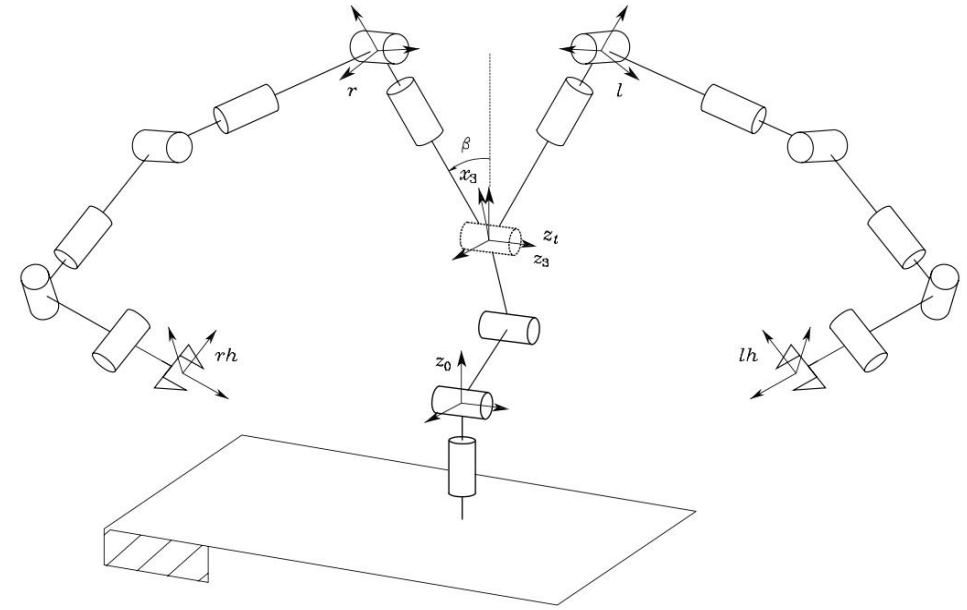
- Actuator
 - An actuator is a device that converts energy into mechanical motion, enabling robots to perform physical tasks.
 - Essentially, it's the "muscle" of the robot, translating commands from the control system into movement or force. How to generate actions?



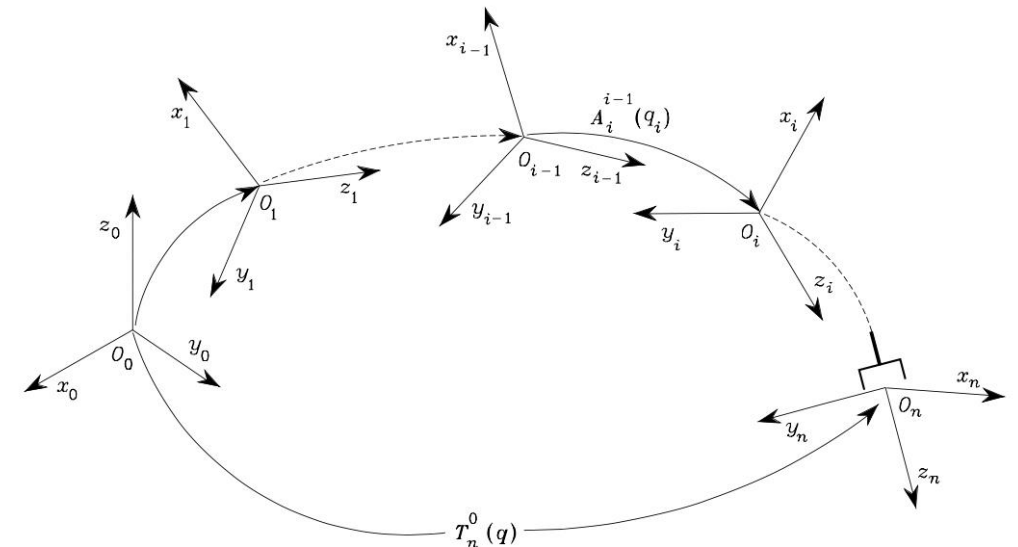
Composing Transforms

- An agent capable of manipulation
 - Simply a number of links connected via joints.
- Each link is associated with a coordinate frame.
- The coordinate frames can be composed.

Physical agent



Kinematic chains



Defining a robot

- A robot consists of a series of rigid links, connected by joints (motors):
 - **Joints provide relative motion**
 - **Links are rigid bodies between joints**
- Each joint provides one degree-of-freedom (sometimes joints may have a coupling).



Universal Robot Description Format (URDF)

What: Unified Robot Description Format

Includes kinematic and basic physics description of a robot

How:

- XML format
- Tags: link or joint
- Kinematic tree structure
- Order in the file does not matter



URDF: Describing a Link

```
<link name="base_link">  
  <visual>  
    <geometry>  
      <cylinder length="0.6" radius="0.2"/>  
    </geometry>  
  </visual>  
</link>
```

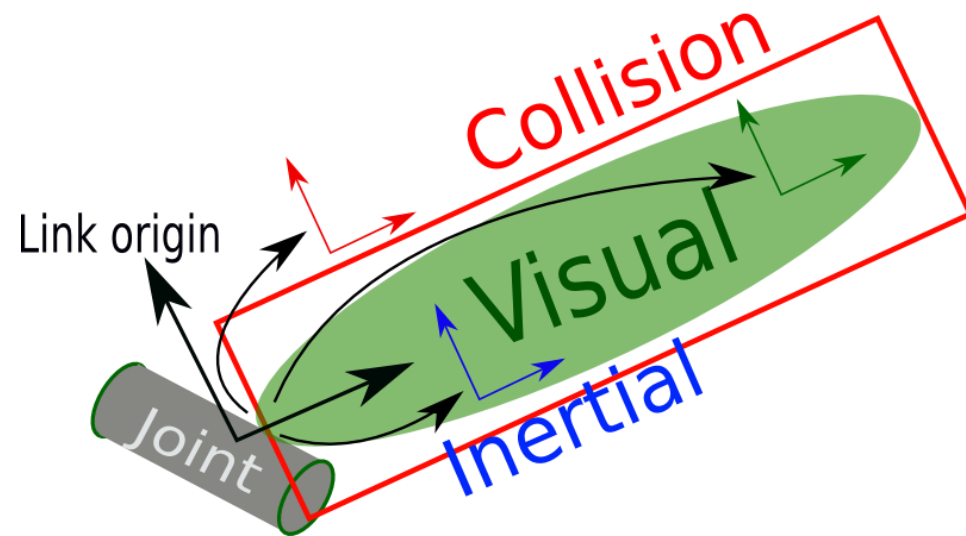
URDF: Describing a Link

```
<link name="base_link">  
  <visual>  
    <geometry>  
      <cylinder length="0.6" radius="0.2"/>  
    </geometry>  
  </visual>  
</link>
```

The 3D geometry of the link can be replaced by other shapes (box, cylinder, sphere) or other mesh files (e.g. STL, OBJ).

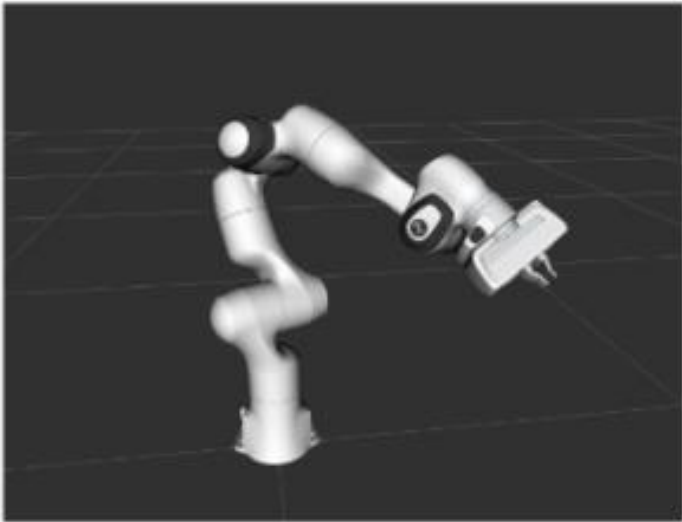
URDF: Physics and collision

```
<link name="base_link">
  <visual>
    <geometry>
      <cylinder length="0.6" radius="0.2"/>
    </geometry>
  </visual>
  <collision>
    <geometry>
      <cylinder length="0.6" radius="0.2"/>
    </geometry>
  </collision>
</link>
```

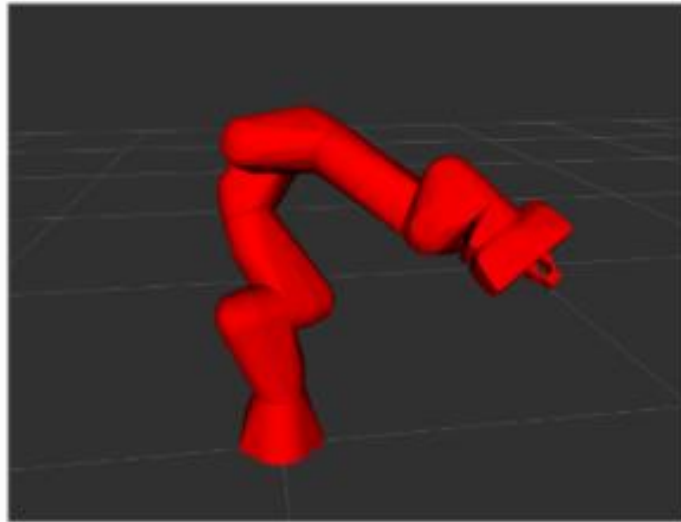


Collision models can be defined at various levels of resolution as required by the decision-making task.

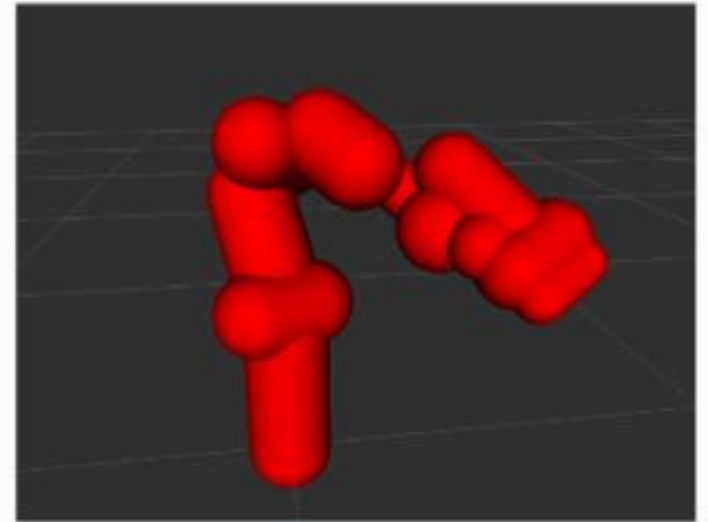
Visual



Collision (Fine)



Collision (Coarse)



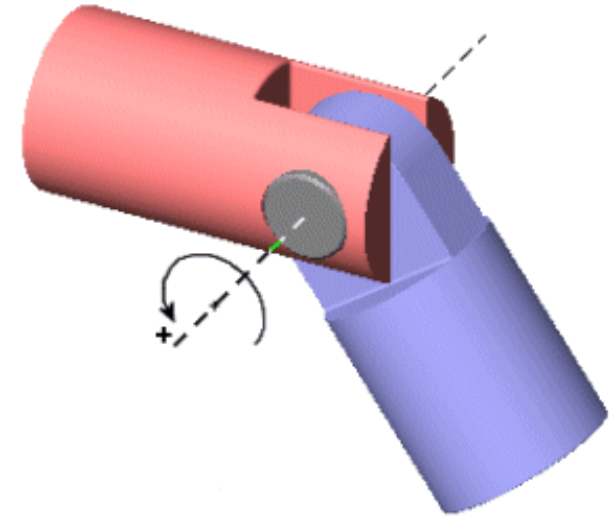
URDF: Describing a Joint

```
<joint name="joint1" type="revolute">  
  <parent link="base_link"/>  
  <child link="right_leg"/>  
    <origin xyz="0 -0.22 0.25"/>  
    <axis xyz="0 0 1"/>  
</joint>
```

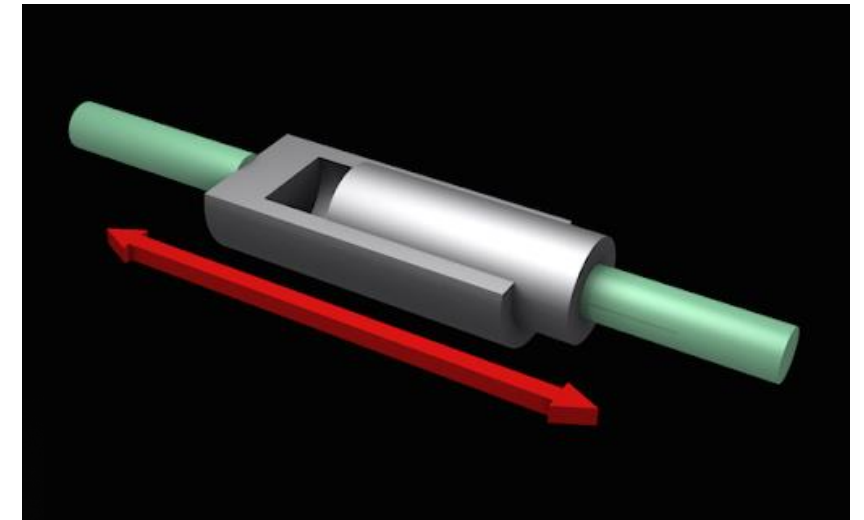
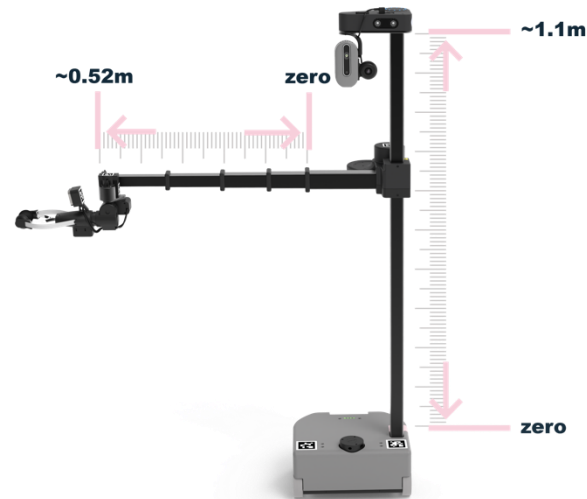
Joint type: fixed, revolute, planar ...

Joints

- Revolute Joint: Single degree of freedom is rotation about an axis.
- Prismatic joint: Single degree of freedom is translation along an axis.



Revolute Joint



Prismatic Joint

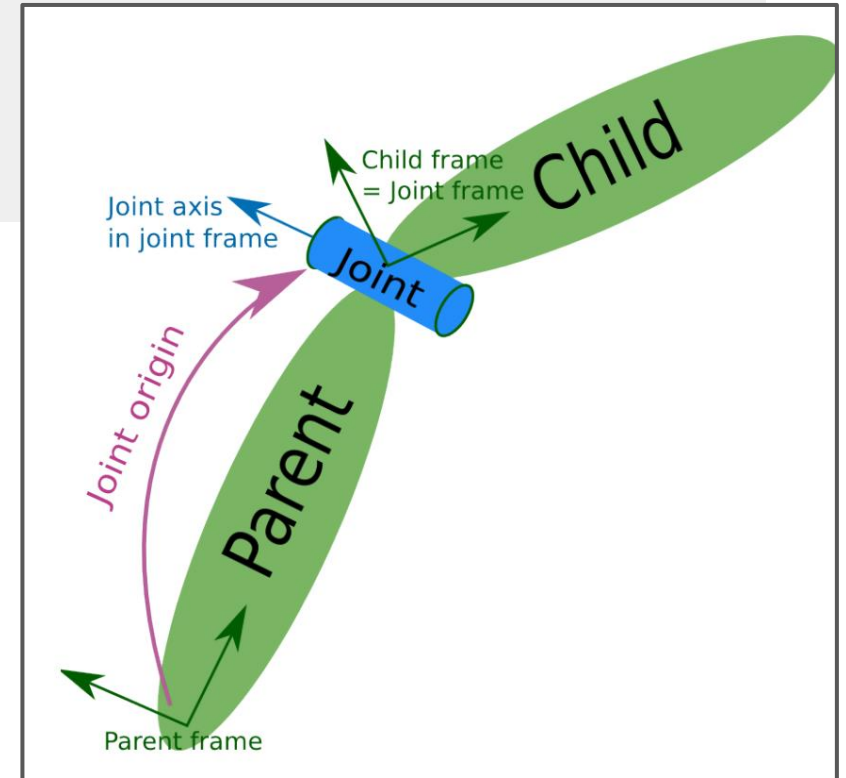
URDF: Describing a Joint

```
<joint name="joint1" type="revolute">  
  <parent link="base_link"/>  
  <child link="right_leg"/>  
    <origin xyz="0 -0.22 0.25"/>  
    <axis xyz="0 0 1"/>  
</joint>
```

Define parent and child links.

Origin: in parent reference frame

Axis: for revolute rotation axis (in joint reference frame)



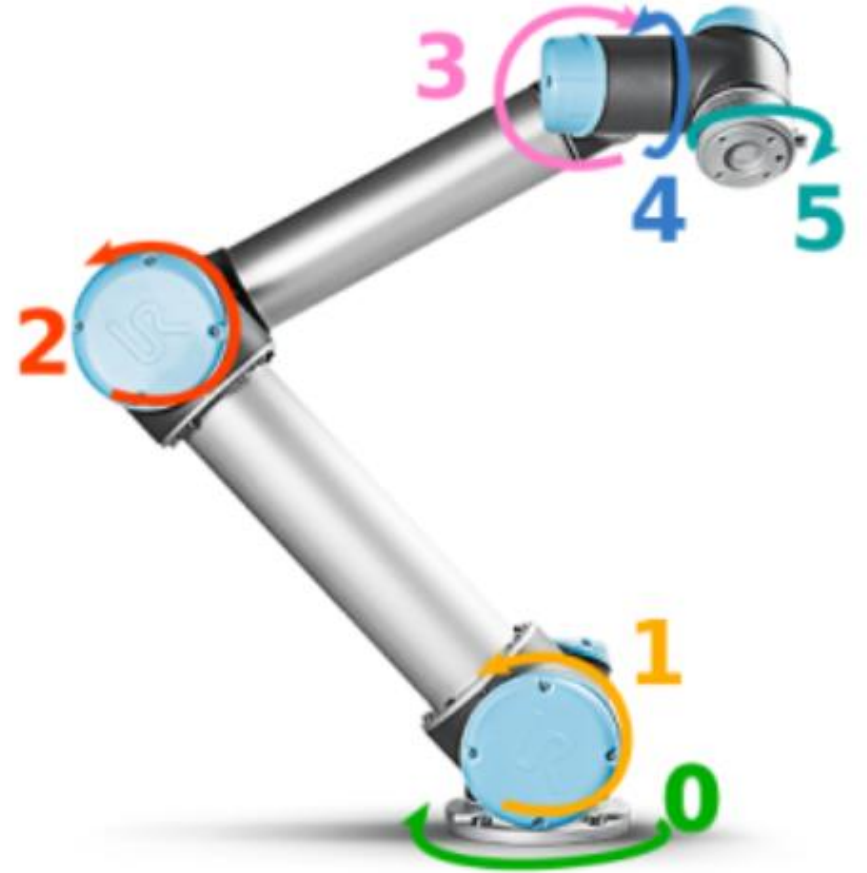
Example: UR5 Arms

Robot characteristics:

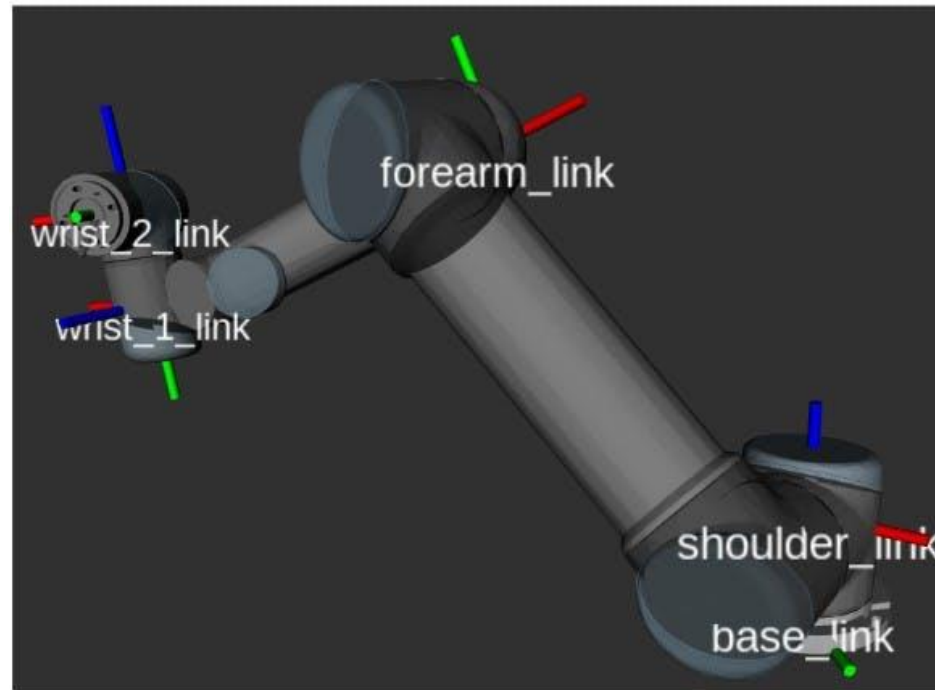
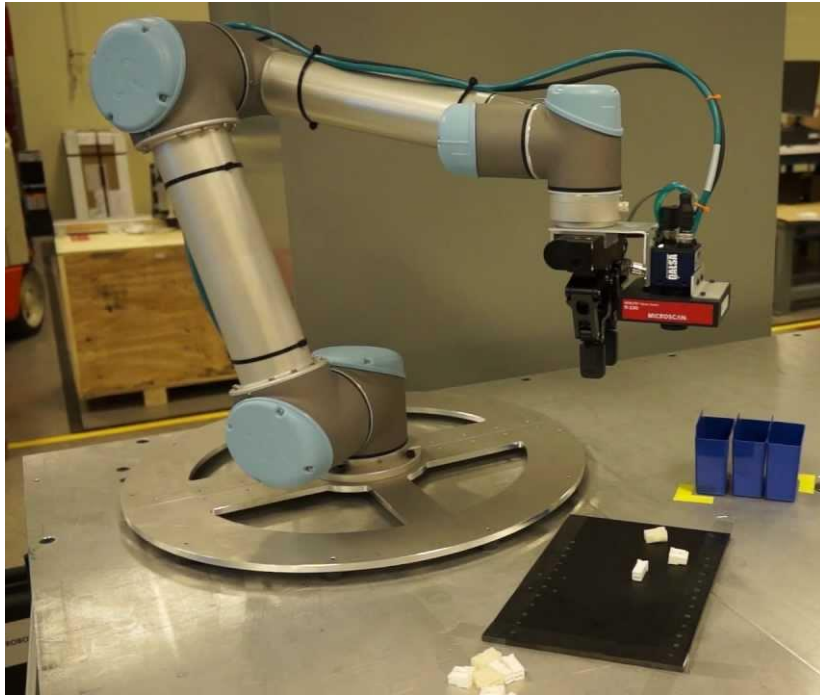
- Serial manipulator

6 DOF URDF:

- One file
- 7 link + 6 revolute joint description



Example of coordinate frames

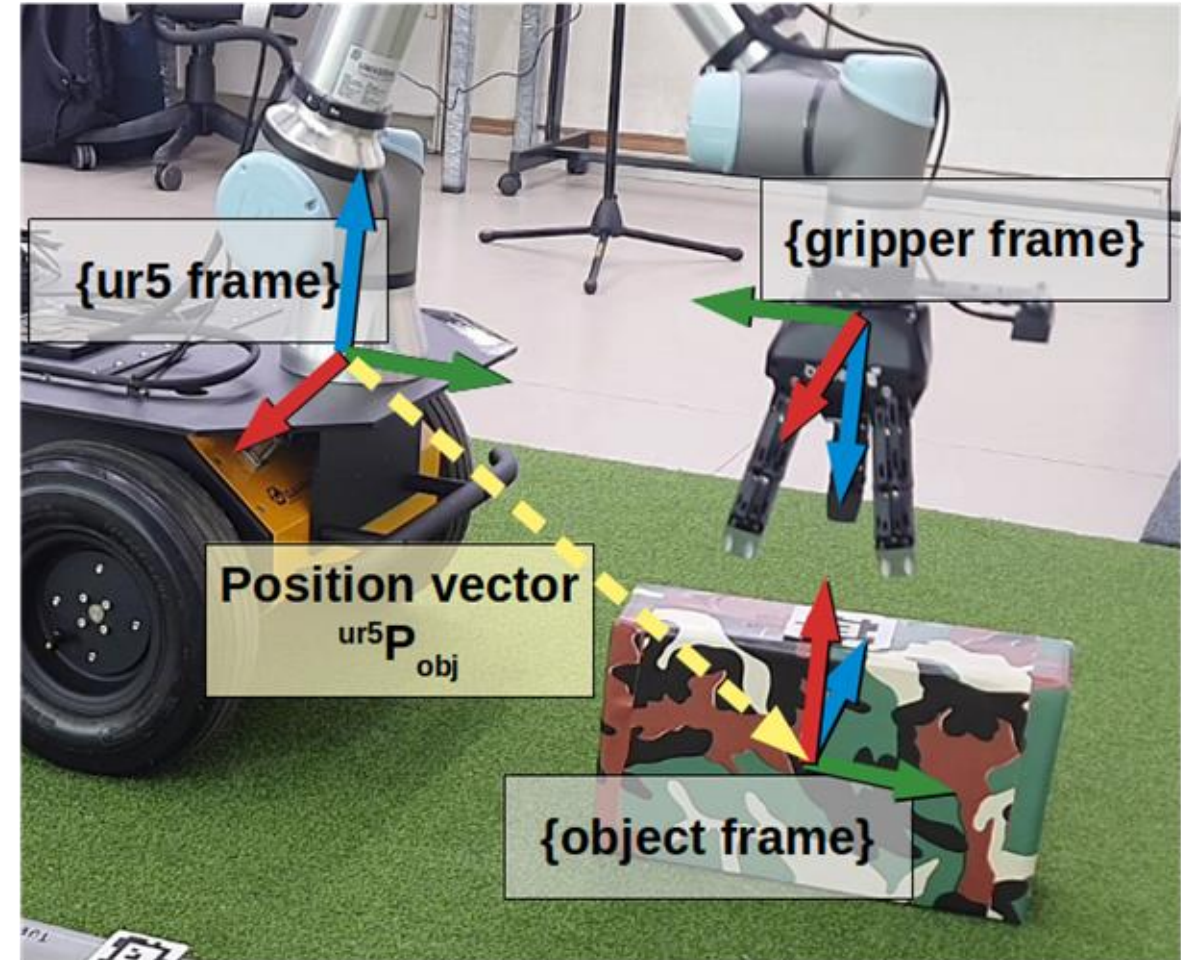
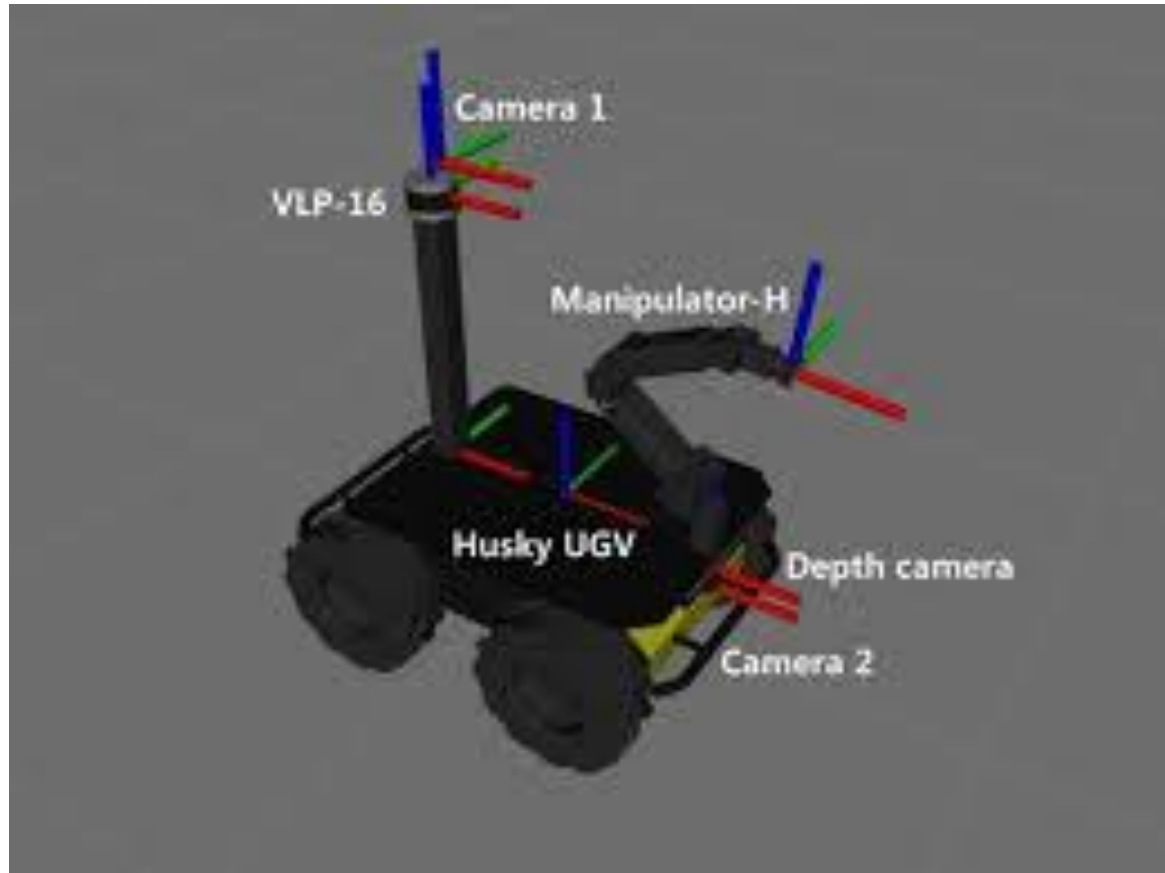


rviz

```
<robot name="ur5">
  <link name = "base_link">
    ...
  </link>
  <joint name = "joint1">
    ...
  </joint>
  ...
  <link name = "wrist_1_link">
    ...
  </link>
  <joint name = "joint_n">
    ...
  </joint>
</robot>
```

URDF

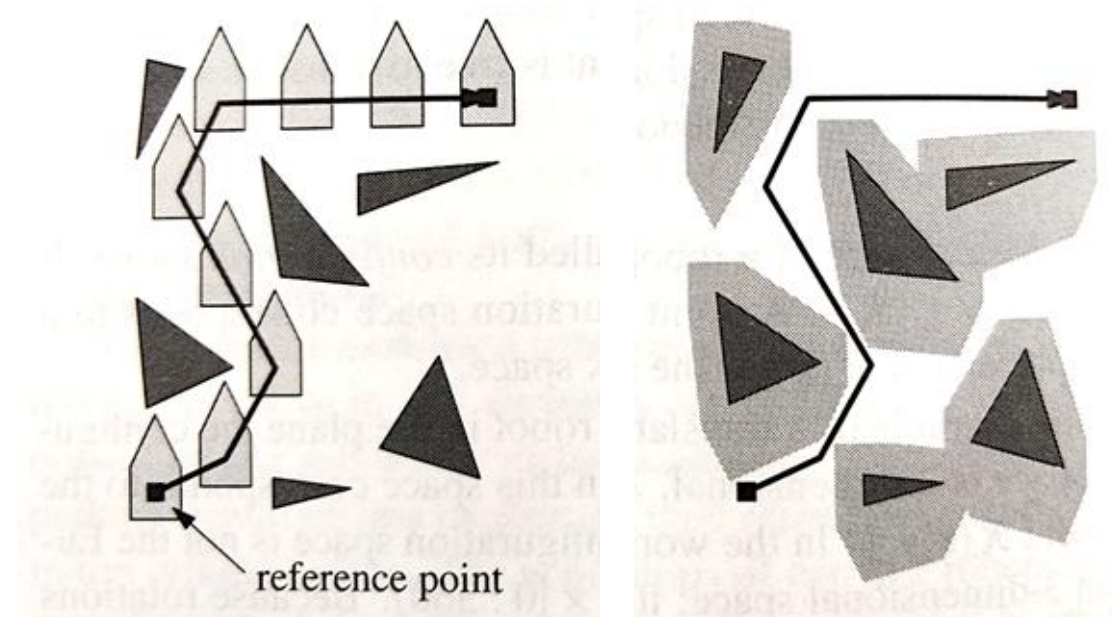
Example of coordinate frames



Task Space and Configuration Space

Central question: which space should the agent plan its actions?

- Task Space
 - Workspace in which the agent operates.
 - May be populated with obstacles.
- Configuration space (C-space)
 - Parameter space of the robot.
 - Space spanned by its allowable degrees of freedom.
- C-space obstacles
 - Non allowable configurations in the C-space.
 - Includes all configurations where the robot collides with the obstacle.



Task space (left) and configuration space (right) for a translating planar agent (cannot rotate).

Task Space and Configuration Space

Task space

- Easier to describe the task.
- Problem: Not all points in the task space may be reachable
- There are physical constraints of the agent's embodiment.

Configuration space (C-space)

- Each point satisfies the intrinsic physical constraints.
- Each point is attainable as it corresponds to a valid configuration of the agent.

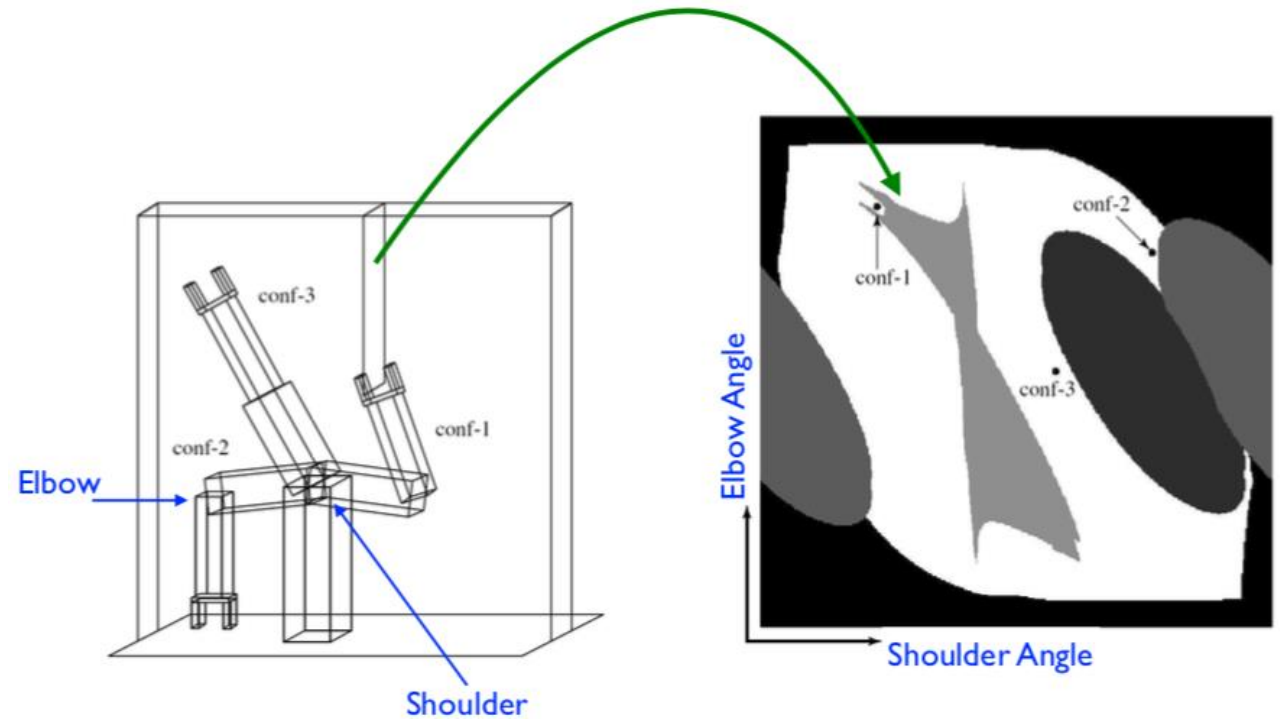
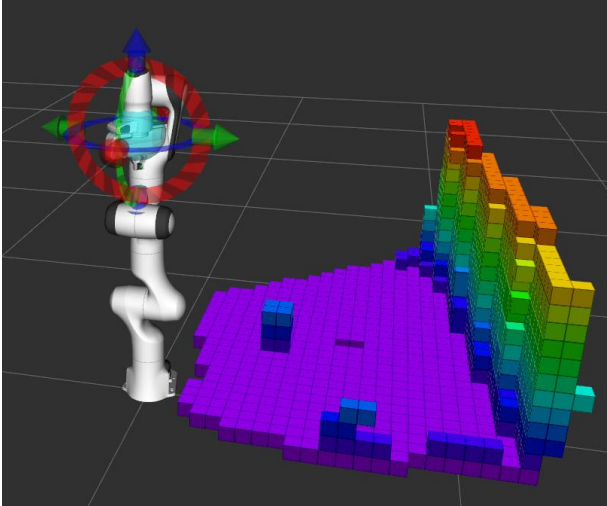


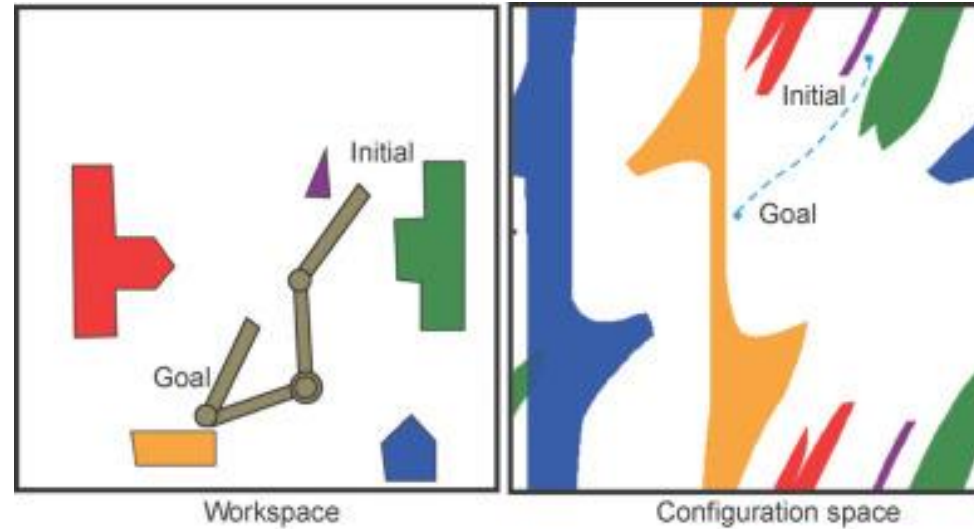
Figure: AIMA Ch 25.4

Example: Configuration space visualization

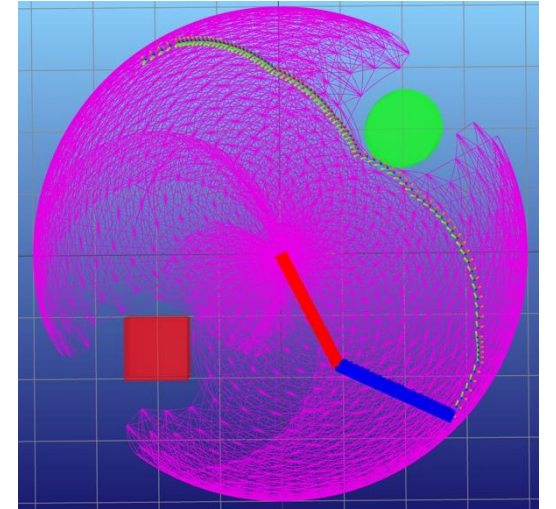
Not all space may be free.



Configuration space and path.



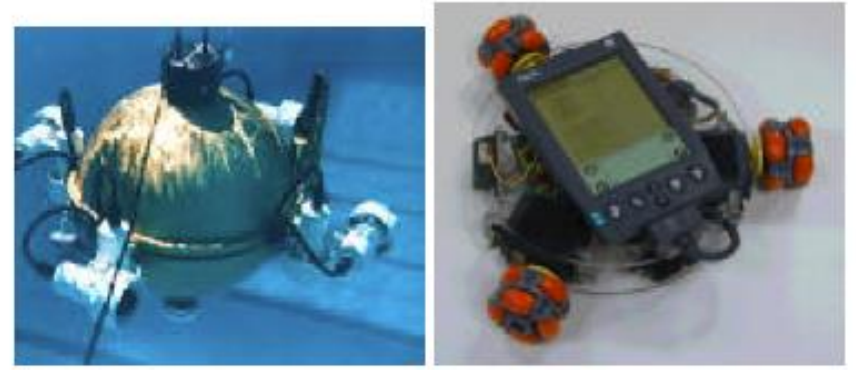
Configuration space and path.



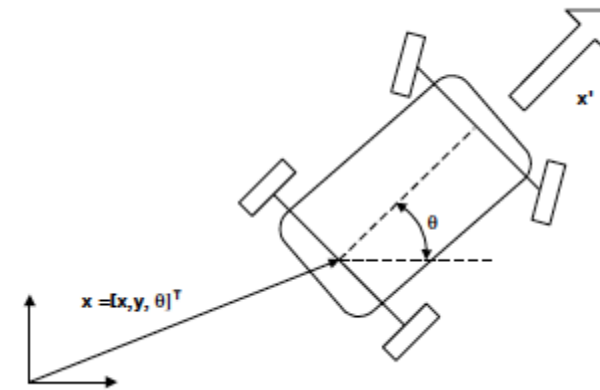
A related concept: *Holonomicity*

- Robot specified in terms of parameters (degree of freedom).
- Holonomic
 - if the number of local degrees of freedom of movement equals the number of global degrees of freedom.
- Non-holonomic otherwise.

Note: even when the robot "has" the flexibility to move in a certain way, its constraints may not permit the entire range of motion



Examples of holonomic vehicles.



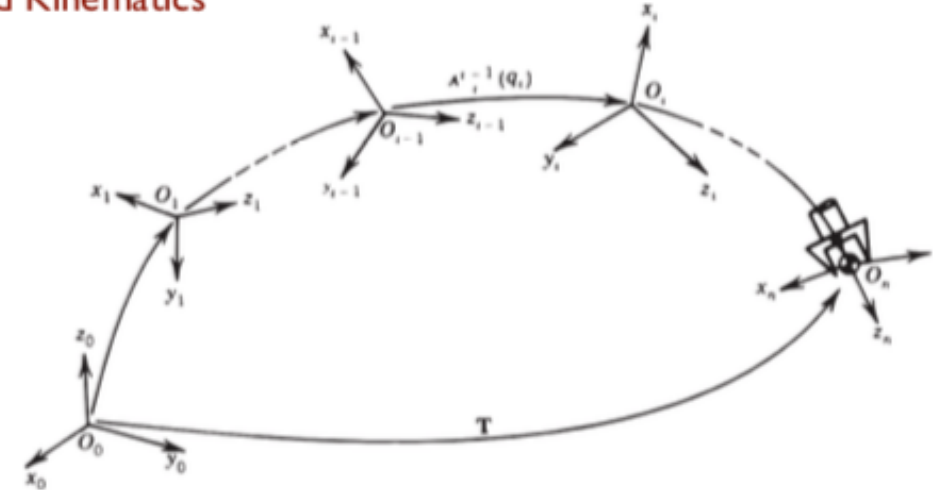
Non-holonomic (car cannot turn in place).

From Configuration Space to Task Space

- Forward kinematics
 - From the parameters specifying the agent (theta parameters) determine the position of the end-effector.
 - Compose the transformations one by one.



Forward Kinematics



Maps configuration space to work space

$$T = T_1^0(\theta_1) T_2^1(\theta_2) \dots T_{n-1}^{n-2}(\theta_{n-1}) T_n^{n-1}(\theta_n)$$

$$= \begin{bmatrix} r_{11} & r_{21} & r_{31} & \Delta x \\ r_{12} & r_{22} & r_{32} & \Delta y \\ r_{13} & r_{23} & r_{33} & \Delta z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x = f(\theta) = f(\theta_1, \theta_2, \dots, \theta_{n-1}, \theta_n)$$

From Task Space to Configuration Space

- Inverse kinematics
 - Determining the setting of the parameters to yield the required end effector position.
 - Given the final position of the end-effector (end point of the arm), determine the theta parameters for the arm.

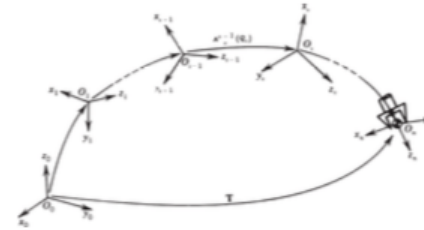


From Task Space to Configuration Space

- Inverse kinematics
 - Solve the inverse problem to obtain theta
- The function f is often non-linear.



Forward Kinematics



$$x = f(\theta)$$

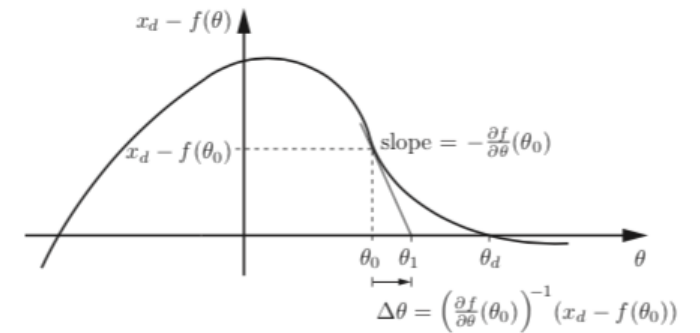
Maps configuration space to work space

Inverse Kinematics

Solve for θ_d in:

$$x_d - f(\theta_d) = 0$$

Typically done using Newton-Raphson method.

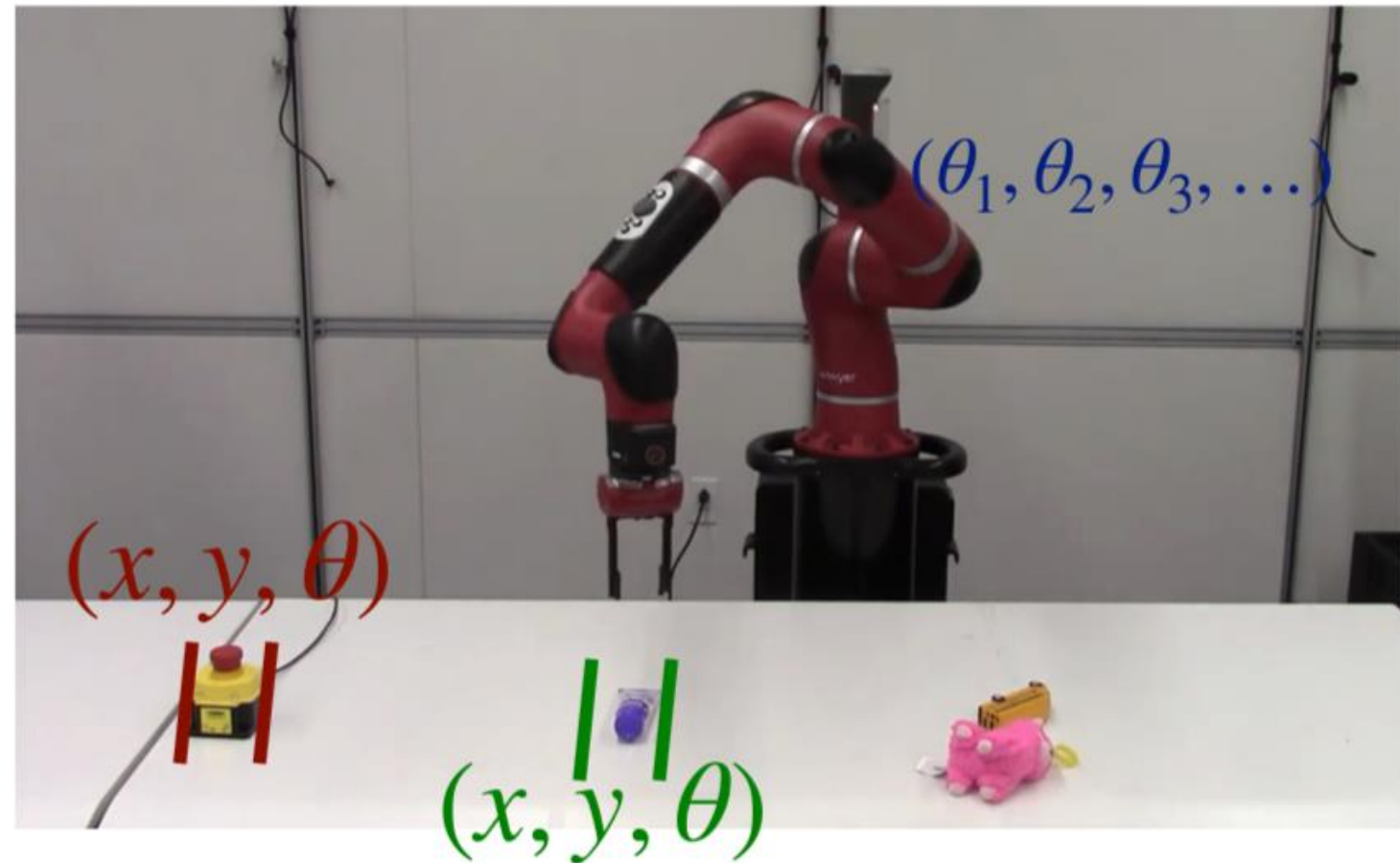


Find configuration(s) that map to a given work space point

Example: Generating Goal-directed Actions

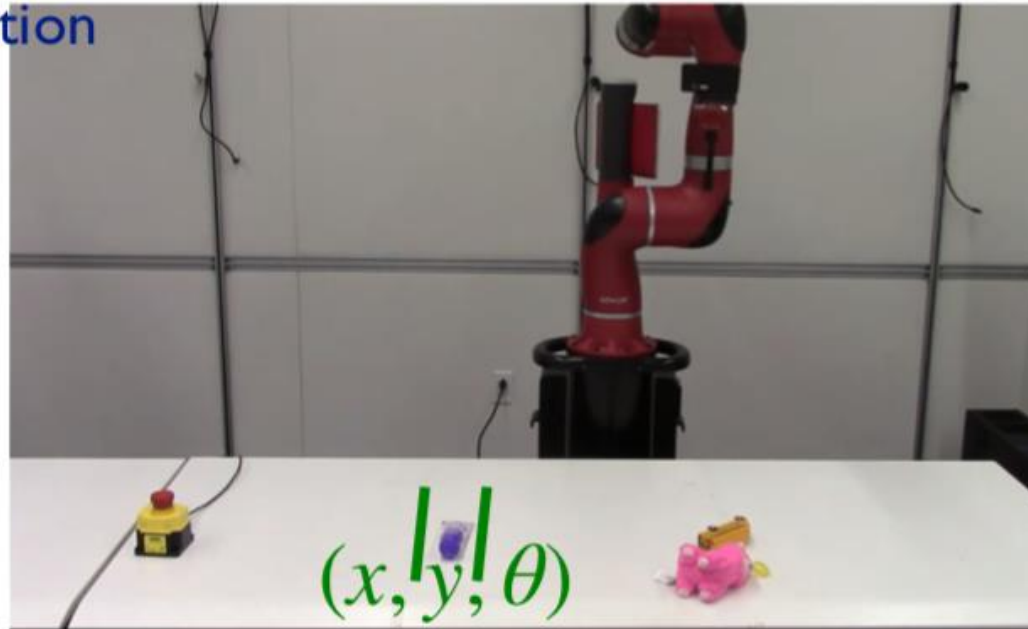
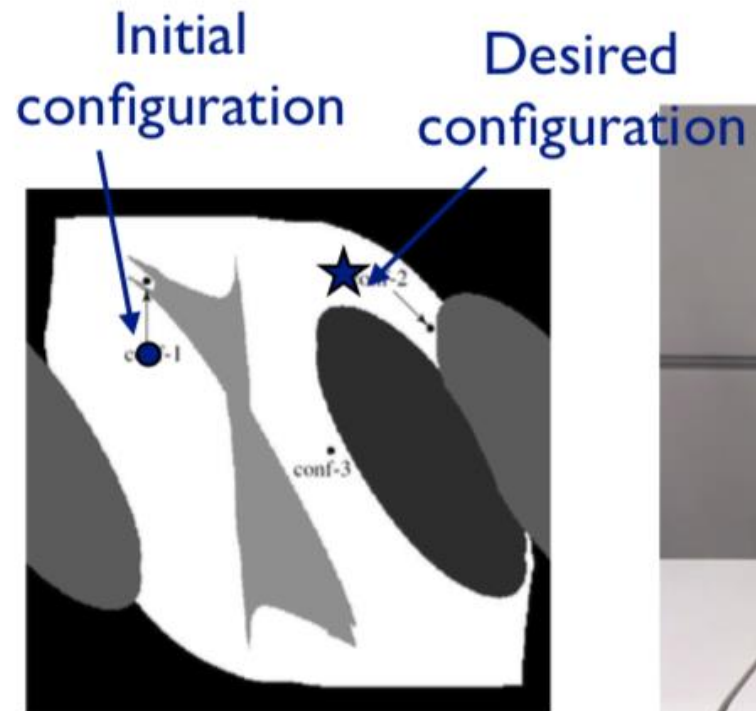
Task

- Given
 - Initial state
 - Goal state
- How to generate actions?



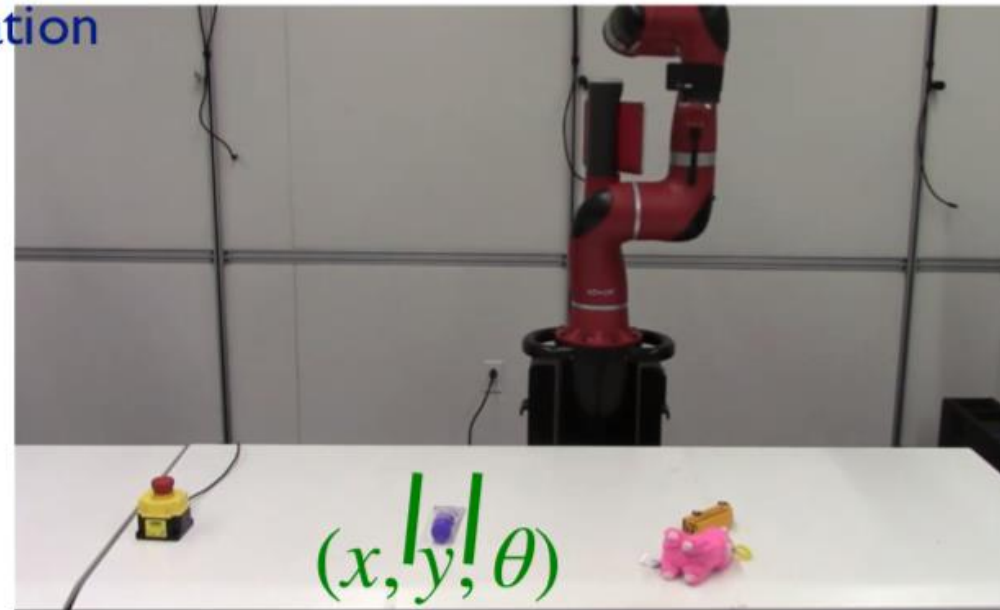
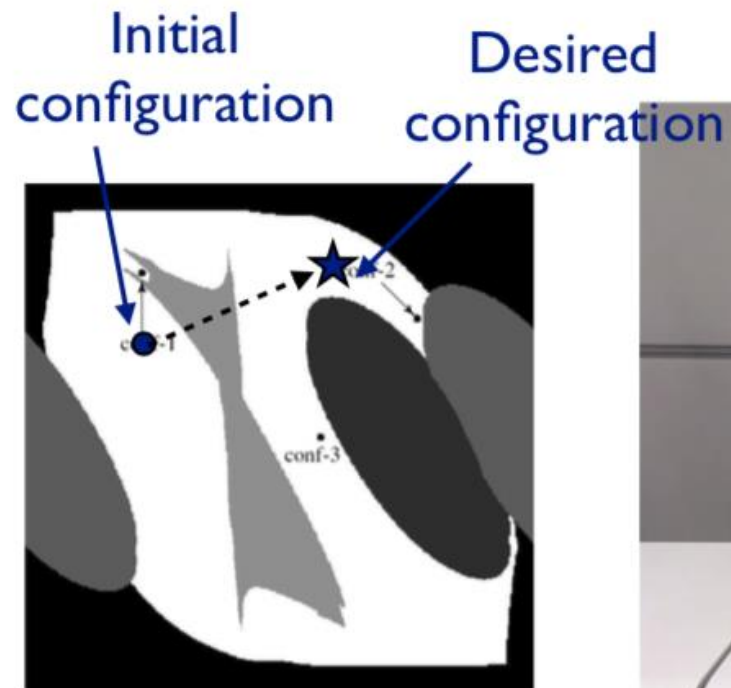
Goal Configuration

I. Task space to Configuration space



Finding a feasible path

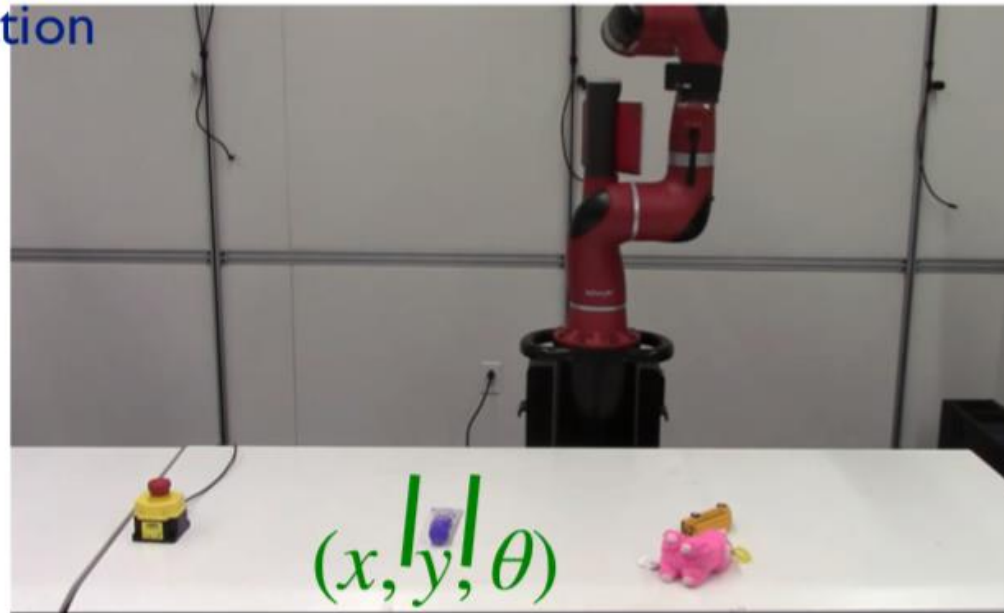
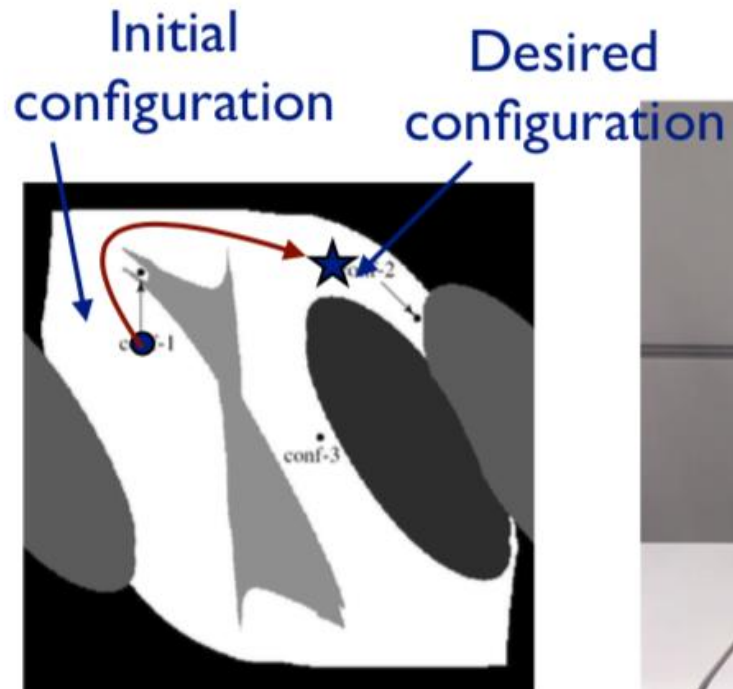
1. Task space to Configuration space
2. Configuration space trajectory



Finding a feasible path

1. Task space to Configuration space

2. Configuration space trajectory

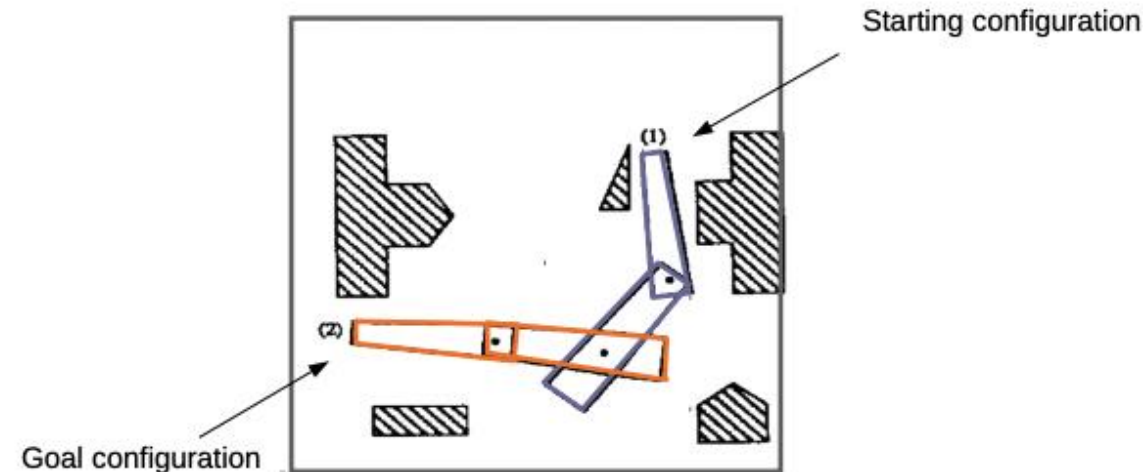


Finding a path – Motion Planning

Given:

- A point-robot (robot is a point in space)
- A start and goal configuration

Find: Path from start to goal that does not result in a collision



Assumptions:

- The position of the robot can always be measured perfectly
- The motion of the robot can always be controlled perfectly
- The robot can move in any direction instantaneously

Modeling Uncertainty

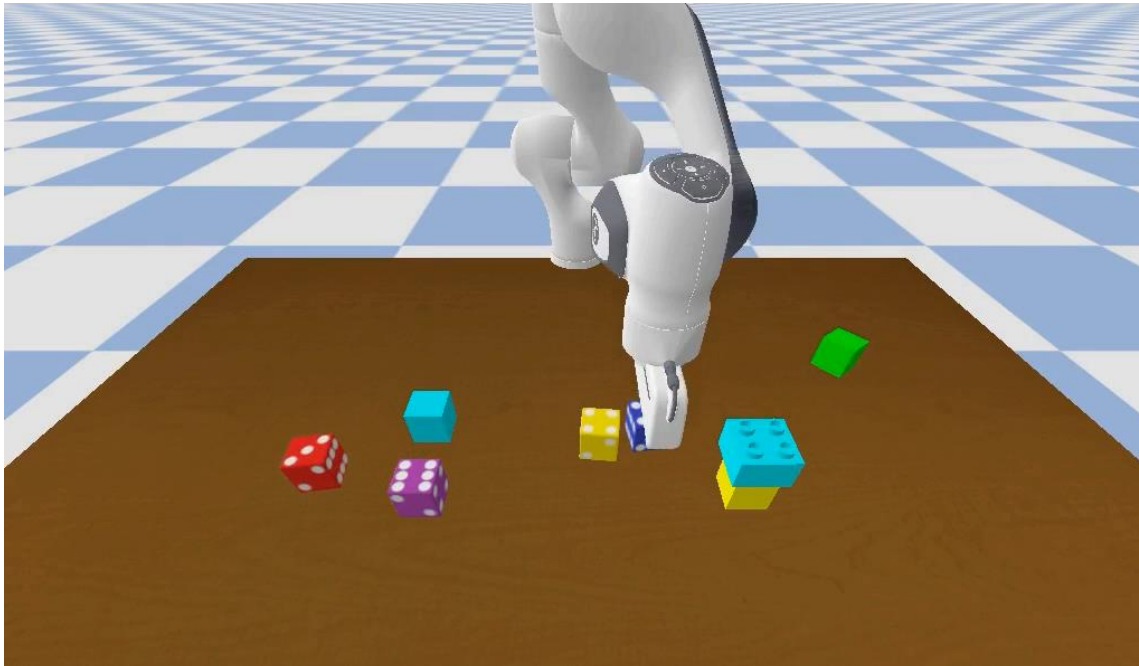
- Action execution always have uncertainty in the real world.
 - Even if the agent "intends" a correct action, it may not be executed perfectly.
 - Note: different from a pure chess playing agent.
- Several sources of uncertainty
 - Consider an autonomous system (a drone, car, Alexa etc.)
 - If the vehicle runs its propulsion system (or motors), what will happen? Is the vehicle working or not?
 - The world is stochastic.
 - Where is the vehicle?
 - Noisy observations.
 - What is around the vehicle?
 - The world is partially unobservable.

Uncertainty arises from action execution and sensing. Any decision-making model must tackle the uncertainty and complexity as above. We will need to develop approaches to model and reason with uncertainty.

Action Uncertainty

Problem: Actions may not have their desired effects. Errors, alternative outcomes, unmodeled dynamics

Solution: Need to model this behaviour using probabilities expressed as the transition model.



Namasivayam et al. ICRA '23, IROS '24



Field experiment where a robot is asked to move to an observation point.

Key Takeaways

- Notion of an agent
 - Interacts with the environment, decide actions and then execute them.
 - Our goal is to develop algorithms that allow goal-directed behaviour to emerge naturally.
- Physical agent representation
 - Notion of a state.
 - Ability to create actuation that leads to movement and physical interaction.
- Space in which to plan actions
 - Task space and the joint space.
 - Essential for planning to realize a behaviour.
- Uncertainty exists
 - Actions taken by the agent are uncertain. Need ways to model and reason with such uncertainty.